Imp Questions for Midterm Mth(621) real analysis

From Past Papers

(i) Define Power Series. .2

Fatima Ali

Sol

An Infinite series of the form

$$\sum_{n=0}^{\infty} a_n (x - x_0)^n$$

where x_0 and $a_0, a_1, \ldots$ are constant is called power series in $x - x_0$.

If $x_0 = 0$ then Power Series becomes

$$\sum_{n=0}^{\infty} a_n x^n.$$

***

Regards to Fatima Ali

(ii) Discuss the series converges or diverges $\{n/2 + 1/n\}$. .3

Sol

The sequence $\{n/2 + 1/n\}$ diverges to ∞, Since if a is any real number, Then

$$\frac{n}{2} + \frac{1}{n} > a \quad \text{if } n \geq 2a$$

Therefore,

$$\lim_{n \to \infty} \left(\frac{n}{2} + \frac{1}{n}\right) = \infty$$

Day: Date:

L.H.S

$= \lim_{n\to\infty} \left(\frac{n}{2} + \frac{1}{n}\right)$

$= \left(\frac{\infty}{2} + \frac{1}{\infty}\right) = (\infty + 0) \quad \{\because \frac{a}{\infty} = 0\}$

$= \infty = R.H.S$

Since the series is divergent.

iii) .2

Discuss the series converges or diverges. $\{n - n^2\}$.

Fatima Ali

Sol

The sequence $\{n-n^2\}$ diverges to $-\infty$. Since if a is real number then

rearranging the sequence

$-n^2+n = -n(n-1)$

$-n(n-1) < a \quad$ if $n > 1+\sqrt{|a|}$

Therefore,

$\lim_{n\to\infty} (-n^2+n) = -\infty$

L.H.S

$= \lim_{n\to\infty} (-n^2+n) = (-(\infty)^2 + \infty)$

$= -\infty^2 + \infty = -\infty = R.H.S$

Hence Sequence is divergent to $-\infty$.

(iv) Bounded Sequence.

Sol A Sequence $\{S_n\}$ is bounded above if there is a real number 'b' such that

$$S_n \leq b \quad \text{for all } n.$$

Bounded below if there is real number 'a' such that

$$S_n \geq a \quad \text{for all } n.$$

we can write it as

$$a \leq S_n \leq b \quad \forall\ n,$$

It is bounded sequence.
Its upper bond is 'b' and lower bound is 'a'.

OR

A sequence is bounded if there is a real number 'r' such that

$$|S_n| \leq r \quad \text{for all } n.$$



Fatima Ali

Related Theorem

v) A Convergent Sequence is bounded.

Day: Date:

Proof

Let

$\lim_{n\to\infty} S_n = S$ is bounded

Sequence.

Let $\varepsilon = 1$ if $\lim_{n\to\infty} S_n = S$

there is an integer N.

Such that

$|S_n - S| < \varepsilon$ if $n \geq N$

$|S_n - S| < 1$ if $n \geq N$

Therefore,

Fatima Ali

$$|S_n| = |(S_n - S) + S|$$
$$\leq |S_n - S| + |S|$$
$$< 1 + |S| \quad \text{if } n \geq N.$$

and

$$|S_n| \leq \max\{|S_0|, |S_n|, \cdots, |S_{N-1}|, 1+|S|\}$$

for all n.

So, $\{S_n\}$ is bounded

Day: Date:

(vi) Define the Number 'e'.

Sol

The Number 'e' is defined by infinite Series and is given

$$e = \sum_{n=0}^{\infty} \frac{1}{n!}$$

Fatima Ali

Where $n! = 1 \cdot 2 \cdot 3 \cdots n$

and "$0! = 1$" ← (Imp) enough define

{Now we expand the series put $n! = 0! \cdot 1! \cdot 2! \cdots n!$}

$$S_n = \frac{1}{0!} + \frac{1}{1!} + \frac{1}{2!} + \cdots + \frac{1}{n!}$$

$\because n! = 1 \cdot 2 \cdot 3 \cdots n$

$$= \frac{1}{1} + \frac{1}{1} + \frac{1}{2} + \cdots + \frac{1}{1 \cdot 2 \cdot 3 \cdots n}$$

$$< 1 + 1 + \frac{1}{2} + \cdots + \frac{1}{(n-1)2^{n-1}} < 3$$

{Just for clearing concept}

if we want to calculate $n!$ then we can write is as

$$n! = (n-1)\,2^{n-1}$$

Let $n = 4$

$$4! = (4-1)2^{4-1}$$

$$4 \cdot 3 \cdot 2 \cdot 1 = 3(2^3) \Rightarrow 24 = 24$$

Day: Date:

(vii) Approximation of Number 'e'.

Sol By the definition of 'e'

$$e = \sum_{n=0}^{\infty} \frac{1}{n!} = \frac{1}{0!} + \frac{1}{1!} + \frac{1}{2!} + \frac{1}{3!} + \cdots$$

Now

Define the Sequence of partial Sums and remainders, for all $n \in N$:

$$S_n = \sum_{n=0}^{\infty} \frac{1}{n!} = \frac{1}{0!} + \frac{1}{1!} + \frac{1}{2!} + \cdots + \frac{1}{n!}$$

Let $R_n = e - S_n = \frac{1}{(n+1)!} + \frac{1}{(n+2)!} + \frac{1}{(n+3)!} + \cdots$

Since all the terms in the Sum are strictly positive, it is clear that $R_n > 0$

Now, we derive an upper bound on an arbitray R_n,

Assume that $n \geq 1$

$$R_n < \frac{1}{(n+1)!} + \frac{1}{(n+1)!(n+1)} + \frac{1}{(n+1)!(n+1)^2} + \cdots$$

$$= \frac{1}{(n+1)!}\left[1 + \frac{1}{n+1} + \frac{1}{(n+1)^2} + \cdots\right]$$

$$= \frac{1}{(n+1)!} \sum_{k=0}^{\infty} \frac{1}{(n+1)^k}$$

Fatima Ali

Day:

$$R_n = \frac{1}{n!n}$$
$$\leq \frac{1}{n!}$$

We have for every $n \geq 1$.

$$0 < R_n < \frac{1}{n!n}$$

adding S_n to the above

$$S_n + 0 < S_n + R_n < S_n + \frac{1}{n!n}$$

Now negate the inequality and add e

Regards to Fatima Ali

$$e - \frac{1}{n!n} < S_n < e$$

* for concept clearing *

{When truncating the infinite series for e after $n+1$ (i.e. the last term is $\frac{1}{n!}$) this partial sum S_n is strictly less than e, but differ from e by no more than $\frac{1}{n!}$}

The algorithms described below is that when both S_n and $S_n + \frac{1}{n!}$ are rounded to the same value, So this is the correct approximation of 'e'

We Conclude that

$$0 < R_n < \frac{1}{n!n}$$

tak $n = 10$, Thus

S_{10} with approximate e with an error less than 10^{-7}

Day: Date:

(viii) Principle of Mathematical induction.

Sol

Let $P(n)$ be a Mathematical Statement, where $n \in N$ (or Z^+) if

- $P(1)$ is true → (Basis Step)
- $P(n)$ is true implies $P(n+1)$ is true (inductive step)

Then

$P(n)$ is true for all $n \in N$ (or Z^+).

For Example

Let

$P(n): 1+2+3+\cdots+n = \frac{n(n+1)}{2}$

→ Basis step:

$P(1)$ is true

Put $n=1$

$1+2+3+\cdots+1 = \frac{1(1+1)}{2}$

imp Inform { the sum of all successive integers starting and ending at same number is equal to that number }

L.H.S = 1 and R.H.S = $\frac{2}{2}$ = 1

Prepared by
Fatima Ali

Day: Date:

Inductive Step:-

for P(n).

Put $n = k$

$$1 + 2 + 3 + \cdots + k = \frac{k(k+1)}{2}$$

Put $n = k+1$

$$1 + 2 + 3 + \cdots + (k+1) = \frac{(k+1)(k+1+1)}{2} \quad -A$$

L.H.S

$$= 1 + 2 + 3 + \cdots + (k+1)$$

$$= \underline{1 + 2 + 3 + \cdots + k} + (k+1)$$

$$= \frac{k(k+1)}{2} + (k+1)$$

$$= \frac{k(k+1) + 2(k+1)}{2}$$

prepared by
Fatima Ali

$$= \frac{(k+1)(k+2)}{2}$$

$$= R.H.S \text{ of } (A)$$

Hence Proved!

Day: Date:

(ix) Prove that $\sqrt{8}$ is an irrational Number.

Sol We Prove it by Contradiction.

prepared by
Fatima Ali

Let $\sqrt{8}$ is a rational number.

$\sqrt{8} = \frac{p}{q}$ ∴ p and q are co-primes and $q \neq 0$.

Now $\sqrt{8} = \sqrt{2 \times 4}$

$= \sqrt{2} \times \sqrt{4}$

$= \pm 2\sqrt{2}$

So ± 2 is a rational number but $\sqrt{2}$ is an irrational number and, Product of rational and irrational number is an irrational number.

So $\sqrt{8}$ is an irrational number. (For 2 Marks Qs)

(For long) →

let $\sqrt{8} = p/q$

squaring both sides, we have

$8 = p^2/q^2$

Day:

$\Rightarrow p^2 = 8q^2$ —(A)

Since p^2 is divided by 8 then $\frac{p^2}{8}$ then $\frac{p}{8}$

So 8 is a factor of p and let

$p = 8a$

Now $p^2 = 64a^2$

Put in A

$64a^2 = 8q^2$

$\frac{64}{8}a^2 = q^2$

$8a^2 = q^2$ —(B)

Since q^2 is multiplied by 8 and q is also Multiple of 8

So 8 is a factor of q.

Since p and q has common factor, So, our assumption of $(\sqrt{8} = p/q)$ is wrong.

Hence $\sqrt{8}$ is an irrational Number.

prepared by
Fatima Ali

Day: Date:

x) Define Supremum and Infimum.

Sol

Supremum

Let a set A, if b is an upper bound of A and no number/element less than b is an upper bound of A.

Then b is the supremum of set A.

$$b = \sup A.$$

fatima ali

Infimum

Let a set A, if "a" is a lower bound of A and no number/element greater than a is a lower bound then, "a" is an infimum of A.

$$a = \inf A.$$

xi) Define Limit Superior and Inferior for S_n

Day: Date:

Sol *Limit Superior*

if a sequence

prepared by
Fatima Ali

$\{S_n\}$ is bounded above and does not diverges to $-\infty$. then there is a unique real number $\bar{S}$, such that, if $\epsilon > 0$,

$$S_n < \bar{S} + \epsilon \quad \text{for large } n.$$

and

$$S_n > \bar{S} - \epsilon \quad \text{for infinitely many } n.$$

Here $\bar{S}$ is called limit superior.

$\longleftrightarrow$

Limit Inferior

if $\{S_n\}$ is bounded below and does not diverges to ∞, then there is a unique real number $\underline{S}$ such that, if $\epsilon > 0$

$$S_n > \underline{S} - \epsilon \quad \text{for large } n$$

and $S_n < \underline{S} + \epsilon$ for infinitely many n.

Here $\underline{S}$ is the limit inferior of $\{S_n\}$.

Day: Date:

(xii) Define Cauchy Sequence:

Sol

A sequence $\{S_n\}$ of real number is called a cauchy sequence if for every $\varepsilon > 0$, there is and integer N such that

$$|S_n - S_m| < \varepsilon \text{ if } m, n \geq N.$$

(xiii)

Fatima Ali

Define oscillate Series.

Sol

A divergent series that does not diverges to $\pm\infty$ is said to be oscillate series or oscillatory.

(xiv) Find the limit of the sequence

$$S_n = \frac{1}{n} \sin n\frac{\pi}{4} + \frac{2(1+3/n)}{1+\frac{1}{n}}$$

Sol

$$\lim_{n\to\infty} S_n = \lim_{n\to\infty} \frac{1}{n} \sin n\frac{\pi}{4} + \frac{2(1+3/n)}{1+1/n}$$

$$= \frac{1}{\infty} \sin \frac{\infty(\pi)}{4} + \frac{2(1+\frac{3}{\infty})}{1+1/\infty}$$

$(\because \frac{a}{\infty} = 0)$

$$= 0 + \frac{2(1+0)}{1+0} = 2 \text{ ans}$$

Day: Date:

xv) Find Limit

$$S_n = \lim_{n\to\infty} \frac{(n/2) + \log n}{3n + 4\sqrt{n}}$$

$$= \lim_{n\to\infty} \frac{n[(1/2) + (\log n)/n]}{n(3 + 4n^{-\frac{1}{2}})}$$

$$= \frac{1/2 + (\log\infty)/\infty}{3 + 4\frac{1}{\sqrt{\infty}}}$$

Fatima Ali

$$= \frac{1}{2\cdot 3} = \frac{1}{6} \quad \underline{\text{ans}}$$

xvi)

For the Power Series

$$\sum \frac{\sin n\pi/6}{2^n} (x-1)^n$$

what is a_n, x_0 and R?

Sol

$$\lim_{n\to\infty} \sup |a_n|^{1/n} = \lim_{n\to\infty} \sup \left(\frac{\sin n\pi/6}{2^n}\right)^{1/n}$$

$$= \left(\frac{1}{2^n}\right)^{1/n} \lim_{n\to\infty} \sup \left(|\sin n\pi/6|\right)^{1/n}$$

$$= \frac{1}{2} \left(\sin(\infty)\pi/6\right)^{\frac{1}{\infty}}$$

$$= \frac{1}{2} (\sin\infty)^0$$

$$= \frac{1}{2} (1) = \frac{1}{2}$$

$R = 2$

The series converges absolutely uniformly in closed subinterval of $(-1, 3)$ and diverges if $x < -1$ or $x > 3$.

xvii)

Determine the Radius of convergence for the power series.

$$\sum \frac{x^n}{n!}$$

Sol

$$\lim_{n\to\infty} \left| \frac{a_{n+1}}{a_n} \right| = \lim_{n\to\infty} \frac{n!}{(n+1)!}$$

$$= \lim_{n\to\infty} \frac{n!}{(n+1)n!}$$

$$= \lim_{n\to\infty} \frac{1}{n+1}$$

$$= \frac{1}{\infty+1} = \frac{1}{\infty} = 0$$

Therefore $R = \infty$, that is the series converges for all x.

regard to Fatima Ali

Day: Date:

xviii) Find Radius and Intervals for Power Series

$$\sum n!\, x^n.$$

prepared by
Fatima Ali

Sol

$$\lim_{n\to\infty} \left|\frac{a_{n+1}}{a_n}\right| = \lim_{n\to\infty} \frac{(n+1)!}{n!}$$

$$= \lim_{n\to\infty} \frac{(n+1)\,n!}{n!}$$

$$= \lim_{n\to\infty} (n+1)$$

$$= \infty + 1 \leq \infty$$

Therefore

$$R = 0$$

and the series converges only if $x = 0$.

Day: Date:

(xiv) Determine the series

$$\sum a_n = \sum (-1)^n \left(\frac{n!}{\alpha(\alpha+1)\cdots(\alpha+n-1)} \right), \quad \alpha > 0$$

for absolutely convergent or not ?

Sol

We apply Raabe's test to

$$\sum a_n = \sum \frac{n!}{\alpha(\alpha+1)\cdots(\alpha+n-1)}$$

Since

$\sum |a_n| < \infty$ if $\alpha > 2$

and $\sum |a_n| = \infty$ if $\alpha < 2$.

Therefore

$\sum a_n$ converges absolutely if $\alpha > 2$ (By the definition of absolutely convergent if $\sum |a_n| < \infty$)

But not converges absolutely if $\alpha < 2$.

It doesn't means that $\sum |a_n|$ diverges if $\alpha < 2$.

Fatima Ali

Day _____ Date _____

Prepared by
Fatima Ali

(xy) Cauchy Root Test
State and Proof:-

Sol

Statement

If $a_n \geq 0$ for $n \geq k$,
then

- $\sum a_n < \infty$ if $\lim_{n\to\infty} \sup a_n^{1/n} < 1$.
- $\sum a_n = \infty$ if $\lim_{n\to\infty} \sup a_n^{1/n} = 1$

the test is inconclusive if
$\lim_{n\to\infty} \sup a_n^{1/n} = 1$.

Proof

If $\lim_{n\to\infty} \sup a_n^{1/n} < 1$

there is an r such that

$$0 < r < 1$$

and $a_n^{1/n} < r$ for larg n.

therefor

$$a_n < r^n \quad \text{for larg } n.$$

Since

$$\sum r^n < \infty$$

The comparison tes implies that

$$\sum a_n^n < \infty$$

Day: Date:

If

$$\lim_{n\to\infty} \sup\, a_n^{1/n} > 1$$

$a_n^{1/n} > 1$ for infinitely many n.

So

$$\sum a_n = \infty.$$

prepared by
Fatima Ali

xxi) $\longleftrightarrow$

Cauchy root test is inconclusive if

$$\sum a_n = \sum \frac{1}{n^p}$$

Sol

$$\lim \sup a_n^{1/n} = \lim_{n\to\infty} \left(\frac{1}{n^p}\right)^{1/n}$$

By the property of log exponential

$$= \lim_{n\to\infty} \exp\left(-\frac{p}{n}\log n\right)$$

$\left(\because e^{-p/n \log n}\right)$

$$= 1 \qquad \forall\ p$$

we know that

$$\sum \frac{1}{n^p} < \infty \quad \text{if } p > 1$$

and

$$\sum \frac{1}{n^p} = \infty \quad \text{if } p \le 1.$$

Date:

Best of Luck

Regards to

FATIMA Ali

remember in your precious prayer

Thank you

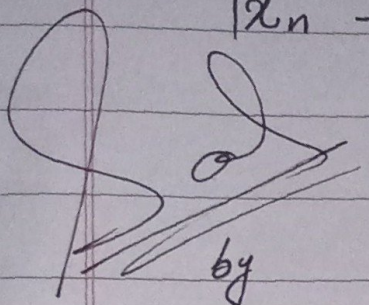
Day: Date:

Suppose that —①

$|x_{n+1} - x_n| \leq \gamma |x_n - x_{n-1}|$, $n \geq 1$

then, Show that —②

$|x_n - x_{n-1}| \leq \gamma |x_1 - x_0|$, if $n \geq 1$.



For $n = 1, 2$ and 3 we find by Solving ①

$n = 1$ in eq ①

$|x_{1+1} - x_1| \leq \gamma |x_1 - x_{1-1}|$

$|x_2 - x_1| \leq \gamma |x_1 - x_0|$ —ⓐ

For $n = 2$ in ①

$|x_{2+1} - x_2| \leq \gamma |x_2 - x_{2-1}|$

$|x_3 - x_2| \leq \gamma |x_2 - x_1|$ —ⓑ

Now look at eq (a) in which $|x_2 - x_1|$ is less than equal to $\gamma |x_1 - x_0|$. Now put it in eq (b) at the place of $|x_2 - x_1|$

So (b) becomes

$|x_3 - x_2| \leq \gamma \cdot \gamma |x_1 - x_0|$

$\leq \gamma^2 |x_1 - x_0|$

it will be continued upto so on.

Now by eq ② for $n = 3$

$|x_3 - x_2| \leq \gamma^{3-1} |x_1 - x_0|$

$\leq \gamma^2 |x_1 - x_0|$ Hence Proved

Day: Date:

Example If $|r| \leq 1$, then

$$\left|\sum_{n=k}^{\infty} r^n\right| = \left|r^k \sum_{n=k}^{\infty} r^{n-k}\right|$$

{Now look at this ↑, obviously, you might be confused that how it comes from}

for clearing concept

let we have a serie

$$\sum_{n=1}^{10} r^n = r^1 + r^2 + r^3 + r^4 + \cdots + r^{10}$$

take common r^1 from the above series we have

$$= r^1\left[r^{1-1} + r^{2-1} + r^{3-1} + \cdots + r^{10-1}\right]$$

It means that r^1 is dessing from each element of above series

Now back to the question ★ ★ ★ Main Solution ★ ★ ★

$$\left|\sum_{n=k}^{\infty} r^n\right| = \left|r^k \sum_{n=k}^{\infty} r^{n-k}\right|$$

if we put $k=0$ in serie Then we have ←

$$= \left|r^k \sum_{n=0}^{\infty} r^n\right|$$

Now the serie will become

$$= \frac{|r|^k}{1-r}$$

if this

$$\frac{|r|^k}{1-r} < \varepsilon$$

then series converges and

$$\left|\sum_{n=k}^{\infty} r^n\right| < \varepsilon \quad \text{if} \quad k \geq K.$$

which implies

$$\lim_{n\to\infty} \sum_{n=k}^{\infty} r^n = 0$$

Solved!